Jacek Karwowski

jacek.karwowski@cs.ox.ac.uk ◇ (+48) 731 77 22 64

EDUCATION

Ph.D. in Computer Science *2022 - now*
University of Oxford
Specialization: Probability theory and category theory

M.Sc. in Pure Mathematics *2018 - 2021*
University of Warsaw
Thesis: Formal semantics of a reversible language

B.Sc. in Computer Science *2018 - 2021*
University of Warsaw
Thesis: Low-latency Apache Parquet library (blogpost)

B.Sc. in Mathematics *2015 - 2018*
University of Warsaw
Thesis: Products in positive opetopic sets

PUBLICATIONS

V. Choudhury, J. Karwowski, and A. Sabry, "Symmetries in reversible programming: from symmetric rig groupoids to reversible programming languages," *Proceedings of the ACM on Programming Languages*, vol. 6, pp. 6:1–6:32, Jan. 2022

ACADEMIC

Class Tutor - Lambda Calculus and Types, University of Oxford Hilary Term 2023

Reviewer - 35th International Conference on Computer Aided Verification 2023

PhD Scholarship - Improving the Long-Term Future, Open Phillantropy 2022 - 2025

Scientific Program Officer - PL in ML Conference 2017

Scientific Lead - Student ML research group, University of Warsaw 2015 - 2016

WORK EXPERIENCE

Researcher - OxAI Safety Labs 2022 - now

OxAI Safety Labs is a research programme intended to introduce students to the technical research on AI safety. I am working with Joar Skalske at Future of Humanity Institute and others on understanding Goodhart's law in RL models (a paper in preparation).

Quant Dev - Xantium 2021 - 2022

Xantium is the algorithmic trading division of Tudor Corporation. I worked on volatility trading, where I built models, implemented execution and managed datasets. Unfortunately, details of my work there are classified under an NDA.

SWE Intern - G-Research 2020

- G-Research is a quantitative research company, developing algorithms for predicting movements in financial markets. I worked on programmers tools for the simulations platform. Unfortunately, details of my work there are classified under an NDA.

Research Intern - Indiana University 2019

· As a part of Global Talent Attraction Program, I conducted research on understanding the semantics of reversible programming languages, in the context of Homotopy Type Theory (paper).

SWE Intern - Microsoft 2019

· I worked on Office365 back-end, adding new features and optimizing the existing ones, and deploying code to production. Details of my work there are classified under an NDA.

Research Intern - Golem Factory GmbH 2017 - 2018

· Golem is a blockchain startup, creating a platform for sharing computational resources using Ethereum network. I developed a probabilistic verification algorithm for programs executed in an untrusted environment, and implemented it for the use-case of hyper-parameter optimization for neural networks (whitepaper, blogpost, another blogpost).

Data Scientist - Applica.ai 2016 - 2017

· Applica is a startup working on ML tools for NLP. I was responsible for creating new models: from reading papers and designing experiments to prototyping models in Python and implementing them on the production cluster.

Student Intern - University College London 2014 - 2015

· Over the academic year, I have been working on a platform for testing high-frequency trading algorithms. Then, during the summer, I worked on a open-source medical data platform.